

# CS & IT ENGINEERING



## Operating System

Memory Management

Lecture -4

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# Recap of Previous Lecture



*doubt ?*

Topic

TLB Mapping

Topic

Segmentation





# Topics to be Covered



**Topic**

**Virtual Memory**

**Topic**

**Effective Memory Access Time**



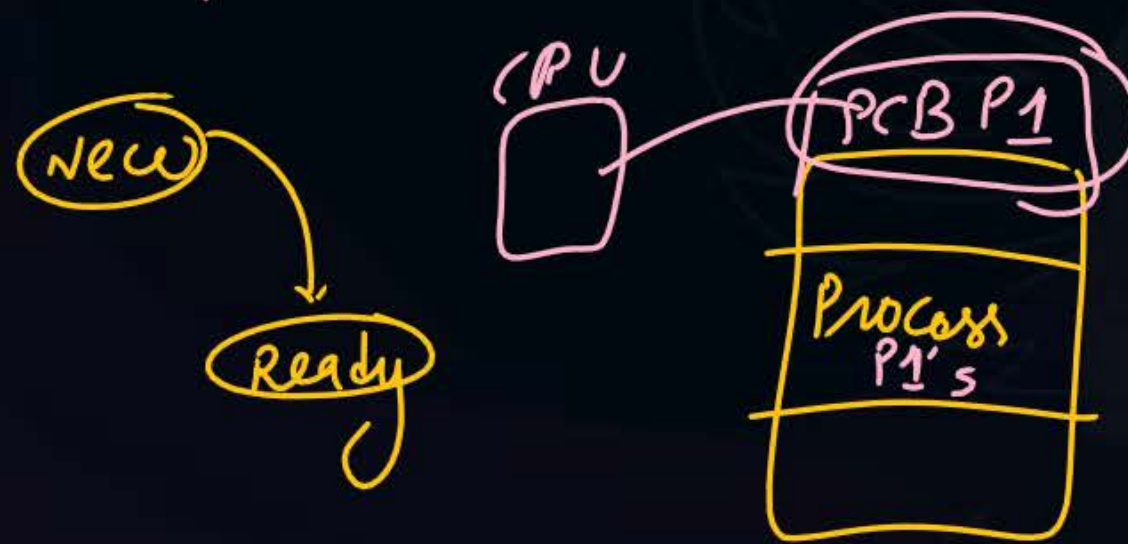


## Topic : Virtual Memory

- Feature of OS
- Enables to run larger process with smaller available memory



keeps only a few pages in  
mm., remaining are kept  
in disk.







# Topic : Virtual Memory

ex:-

Process

Page table

|        |     |               |
|--------|-----|---------------|
| Page 0 | 000 | 10            |
| Page 1 | 001 |               |
| Page 2 | 010 |               |
| Page 3 | 011 | 00            |
| Page 4 | 100 | <del>00</del> |
| Page 5 | 101 | 11            |
| Page 6 | 110 |               |
| Page 7 | 111 | 01            |

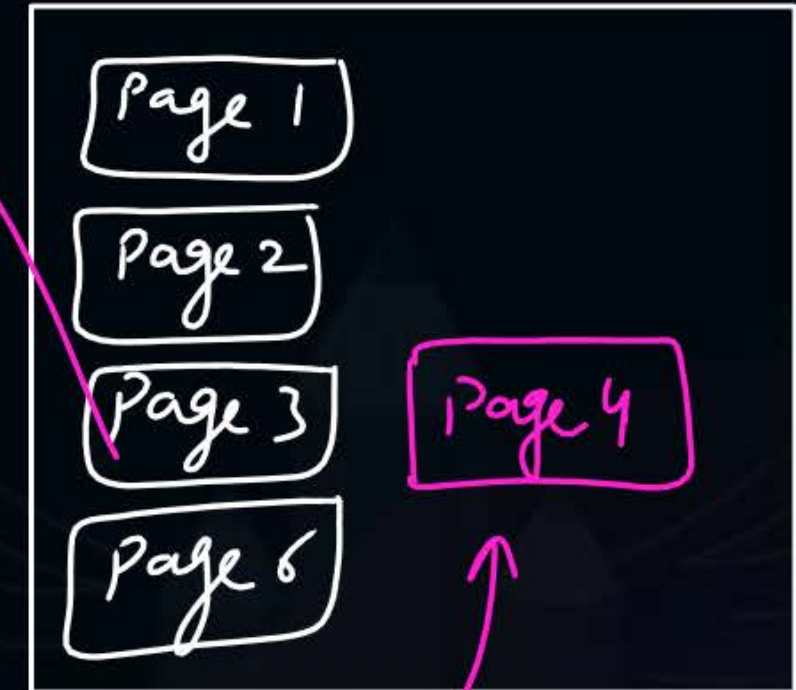
8 pages

Physical mem.

|    |                        |
|----|------------------------|
| 00 | Page <del>4</del><br>3 |
| 01 | Page 7                 |
| 10 | Page 0                 |
| 11 | Page 5                 |

4 frames

Disk



assume CPU generates a L.A. which belongs to page no. 3



search in page table for  $p=3$



Page is not available in m.m. (Page fault)



system call (internal interrupt to OS)



OS provides service of page fault by bringing demanded page into mm and changing page table contents.



The instruction which caused page fault will restart.





## Topic : Demand Paging

- **Demand Paging:**

Bring pages in memory when CPU demands

- **Page Fault:**

When the demanded page is not available in physical memory

- Pure demand paging:-

when process starts then there is no any process page in mm for that process and all the pages are brought to m.m. on demand only.



## Topic : How to Check Page Hit/Fault

|   | frame<br>no  |              |
|---|--------------|--------------|
| 0 | 2            | 1            |
| 1 |              | 0            |
| 2 |              | 0            |
| 3 | 0            | 1            |
| 4 | <del>0</del> | <del>1</del> |
| 5 | 3            | 1            |
| 6 |              | 0            |
| 7 | 1            | 1            |

valid/invalid bit

→ 1  
→ 0

frame no. (page table entry) is valid)  
empty entry (invalid entry)

page 4 replaced by page 3





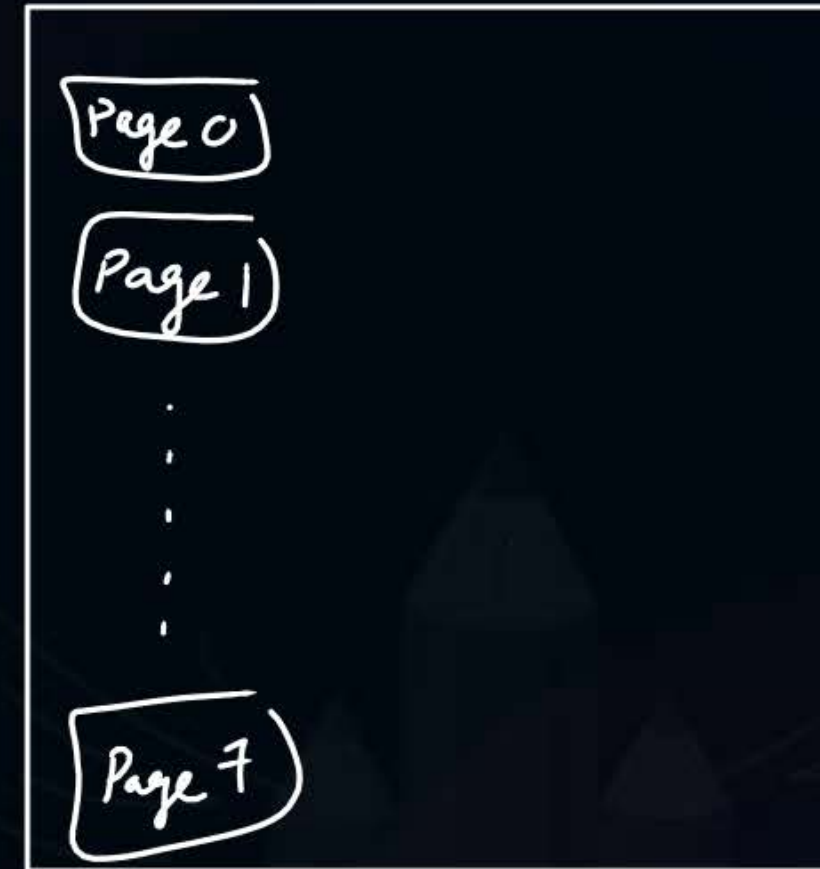
- do not write back non-dirty pages to disk when replaced from m.m.

## Page table

[illegible]

mm

|   |        |
|---|--------|
| 0 | Page 4 |
| 1 | Page 7 |
| 2 | Page 0 |
| 3 | Page 5 |



# Disk

modified  
or  
dirty bit

→ 0  
→ 1

page in mm did not have any write

— 11 — had write done



## Topic : Effective Memory Access Time

Assume Page fault rate =  $p$

CPU generates L.A.

↓  
search in P.T.

Page hit

↓  
P.A.

↓  
Access content from mm using P.A.

page fault

↓  
OS services  
Page fault

E.M.A.T. =

$$(1-p) [t_{mm} + t_{mm}]$$

$$+ p * [t_{mm} + \text{p.f. service time}]$$

or

$$= t_{mm} + (1-p) * t_{mm} + p * \text{p.f.s.t.}$$



CPU generates L.A.

↓  
search in TLB

TLB hit

P.A.

↓  
access mm  
using P.A. to  
get content

TLB miss

search P.T. in mm

Page hit

P.A.

↓  
access mm  
using P.A.  
to get  
content

Page fault

P.f. service

$$E.M.A.T. = H [t_{TLB} + t_{mm}]$$

$$(1-H) * [t_{TLB} + t_{mm} + (1-P) t_{mm} + P * P.f. s.t.]$$



## Topic : Question



TLB hit = 90%

TLB access time = 10 n sec

mm access time = 200 n sec

Page fault rate = 1 %

p.f. service time = 100  $\mu$  sec

E. M. A. T. = 329.8 n sec.

$$= 0.9 * (10 + 200)$$

$$+ 0.1 \left[ 10 + 200 + 0.99 * 200 + 0.01 * 100 * 1000 \right]$$

$$= 329.8 \text{ ns}$$



Page fault service

- dirty page replaced  $\Rightarrow$  P.f.s.t. will be more
- non dirty page replaced  $\Rightarrow$  P.f.s.t. will be lesser



## Topic : Dirty Bit Included (Without TLB)

$$E.M.A.T. = t_{mm} + (1-p) * t_{mm} + p \left[ m * \underset{\substack{\text{for} \\ \text{dirty} \\ \text{pages}}}{P.f.s.T.} + (1-m) * \underset{\substack{\text{for} \\ \text{non-} \\ \text{dirty} \\ \text{pages}}}{P.f.s.T.} \right]$$

$m$  = fraction of pages which are dirty and replaced





## Topic : Question



$$P = 2\%$$

$$M = 5\%$$

$$T_{mm} = 100 \text{ n sec}$$

P. F .S.T. when page dirty =  $200\mu \text{ sec}$

P. F .S.T. when not dirty =  $100\mu \text{ sec}$

E. M. A. T. = \_\_\_\_\_ n sec.

$$E.M.A.T. = 100 + 0.38 * 100$$

$$+ 0.02 * \left[ 0.95 * 100000 + 0.05 * 200000 \right]$$

$$= 2298 \text{ ns}$$



## Topic : Dirty Bit with TLB



$$E.M.A.T. = H * (t_{TLB} + t_{mm})$$

$$+ (1-H) \left[ t_{TLB} + t_{mm} + (1-p) t_{mm} + p \left[ (1-m) \underset{\substack{\text{P.f.s.T.} \\ \text{for} \\ \text{non-dirty}}}{\text{P.f.s.T.}} + m * \underset{\substack{\text{P.f.s.T.} \\ \text{for} \\ \text{dirty}}}{\text{P.f.s.T.}} \right] \right]$$





## Topic : Question

$$H = 85\%$$

$$t_{TLB} = 20 \text{ n sec}$$

$$t_{mm} = 200 \text{ n sec}$$

$$P = 1\%$$

$$M = 2\%$$

$$\text{P.F.S.T. with dirty page} = 100 \mu \text{ sec}$$

$$\text{P.F.S.T. without dirty page} = 50 \mu \text{ sec}$$

$$\text{E. M. A. T.} = \underline{\hspace{2cm}} ?$$

$$\begin{aligned} \text{E.M.A.T.} &= 0.85 * (20 + 200) \\ &+ 0.15 \left[ 20 + 200 + 0.99 * 200 + 0.01 \left[ 0.98 * 50000 + \right. \right. \\ &\quad \left. \left. 0.02 * 100000 \right] \right] \\ &= 326.2 \text{ ns} \end{aligned}$$

P.f.s.t. for non-dirty page = Page transfer time from disk to mm + Page table updation time in mm.

P.f.s.t. for dirty page = page transfer time from mm to disk for replaced dirty page + page transfer time from disk to mm for faulted page + p.t. updation time in mm





## Topic : Question

TLB Hit = 90%

$t_{TLB} = 15 \text{ n sec}$

$T_{mm} = 120 \text{ n sec}$

$P = 2\%$

$M = 1\%$

Time taken to transfer a page b/w mm & sm = 10000 n sec

E.M.A.T. = \_\_\_\_\_?

$$EMAT = 0.9 * (15 + 120)$$

$$+ 0.1 * [15 + 120 + 0.98 * 120$$

$$+ 0.02 [0.99 * 10120 + 0.01 * 20120]]$$

$$= 167.2 \text{ ns}$$

p.f.s.t. for non-dirty pages =  $10000 + 120 = 10120 \text{ ns}$

—||— dirty pages =  $10000 + 10000 + 120 = 20120 \text{ ns}$



## Topic : Question

H.W.

[GATE-2020]



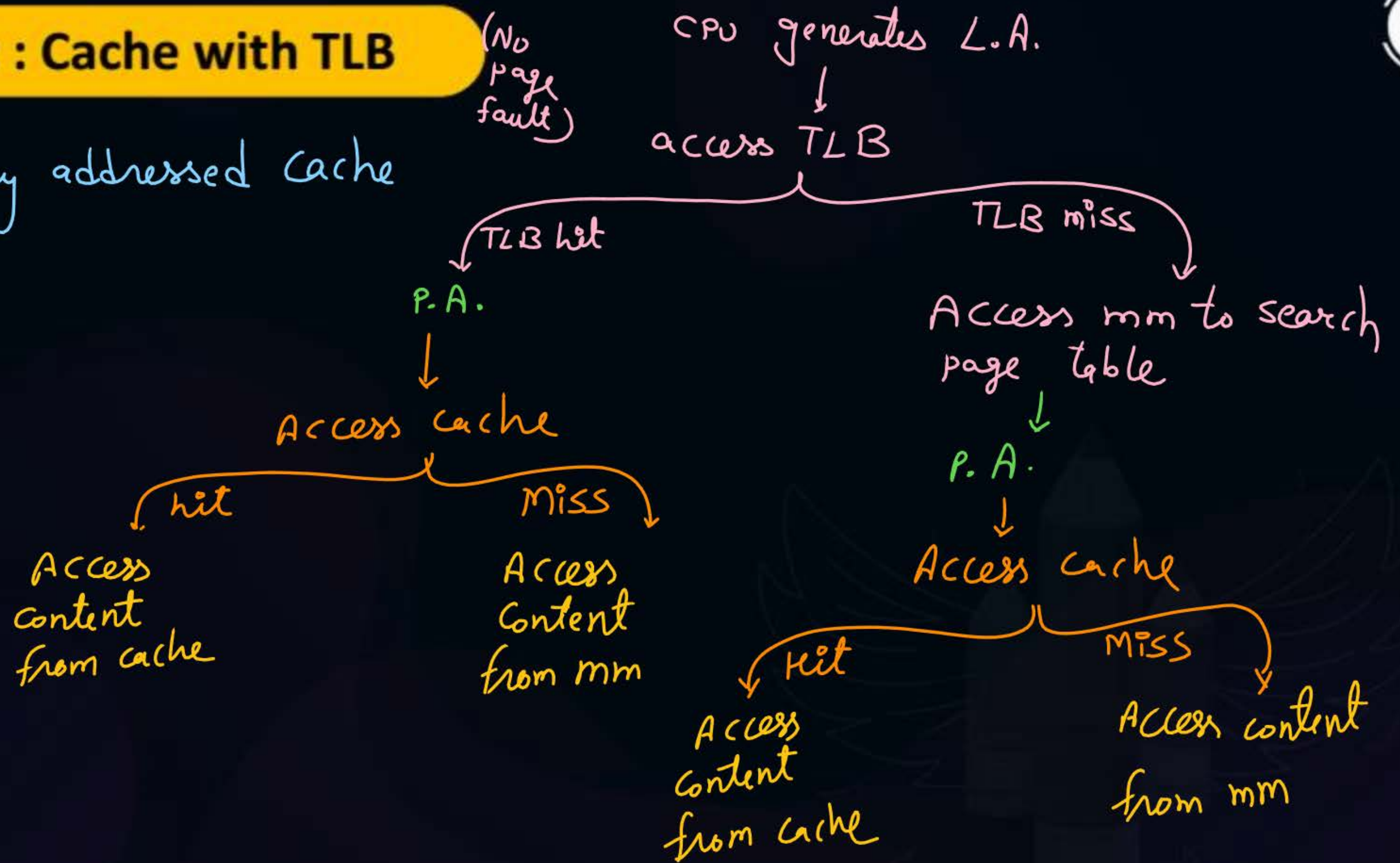
- #Q. Consider a paging system that uses 1-level page table residing in main memory and a TLB for address translation. Each main memory access takes 100 ns and TLB lookup takes 20 ns. Each page transfer to/from the disk takes 5000 ns. Assume that the TLB hit ratio is 95%, page fault rate is 10%. Assume that for 20% of the total page faults, a dirty page has to be written back to disk before the required page is read from disk. TLB update time is negligible. The average memory access time in ns (round off to 1 decimal places) is\_\_\_\_\_.





## Topic : Cache with TLB

Physically addressed cache



$$\text{Content access time using P.A. } (t_{cA}) = H_c (t_{cm}) + (1 - H_c) (t_{cm} + t_{mm})$$

$$\begin{aligned} \text{E.M.A.T.} = & H (t_{TLB} + t_{cA}) \\ & + (1 - H) (t_{TLB} + t_{mm} + t_{cA}) \end{aligned}$$



Ques)  $H_c = 90\%$

$$H_{TLB} = 95\%$$

$$t_{cm} = 20 \text{ ns}$$

$$t_{TLB} = 5 \text{ ns}$$

$$t_{mm} = 200 \text{ ns}$$

$$\begin{aligned} t_{CA} &= 0.9 * 20 + 0.1 * (20 + 200) \\ &= 40 \text{ ns} \end{aligned}$$

$$\begin{aligned} EMAT &= 0.95 * (5 + 40) \\ &\quad + 0.05 * (5 + 200 + 40) \\ &= 55 \text{ ns} \end{aligned}$$



## Topic : Question

H.W.



- #Q. A processor uses 1-level page table for virtual to physical address translation. Page table is stored in the main memory. Further, the processor has a translation look-aside buffer (TLB), with a hit rate of 96%. The TLB caches recently used virtual page numbers and the corresponding physical page numbers. The processor also has a physically addressed cache with a hit rate of 90%. Main memory access time is 10 ns, cache access time is 2 ns, and TLB access time is also 1 ns. Assuming that no page faults occur, the average time taken to access a virtual address is approximately?





## 2 mins Summary

**Topic**

**Virtual Memory**

**Topic**

**Effective Memory Access Time**



**Happy Learning**

**THANK - YOU**

